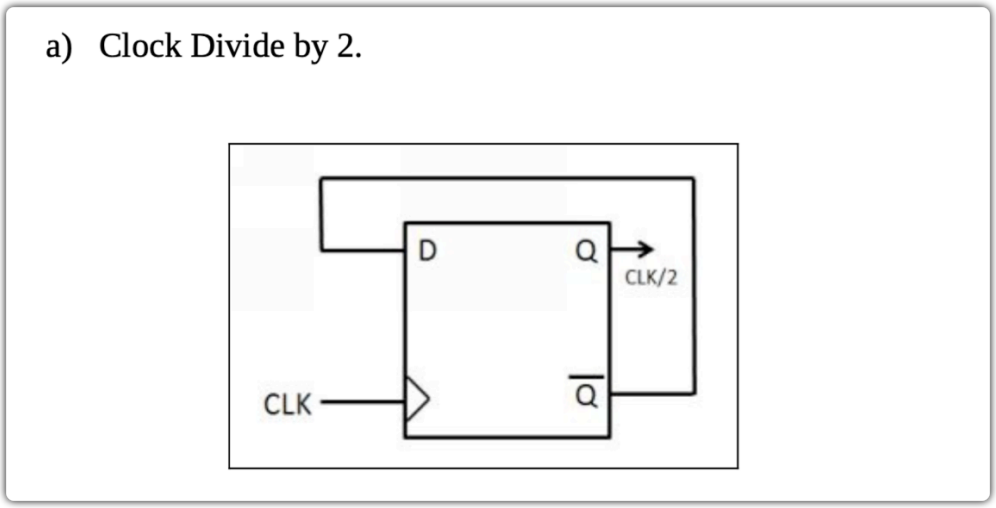
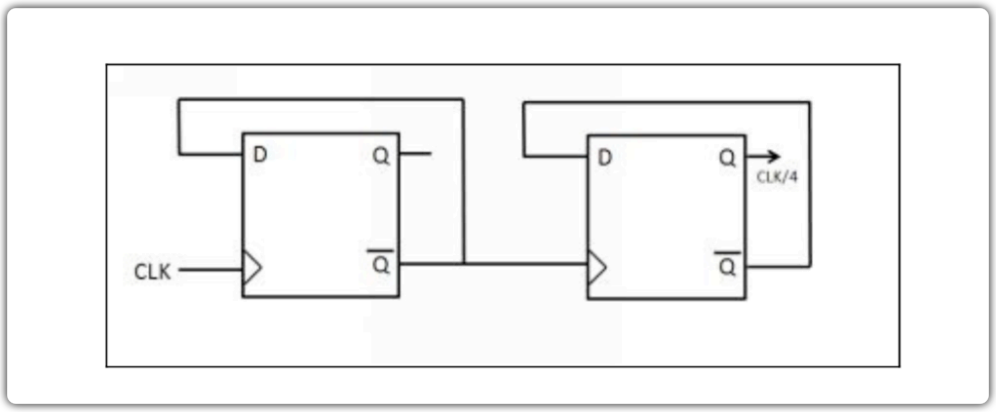
module 02: 偶数分频

1. 偶数分频

- 实现二分频电路，满足50%占空比。



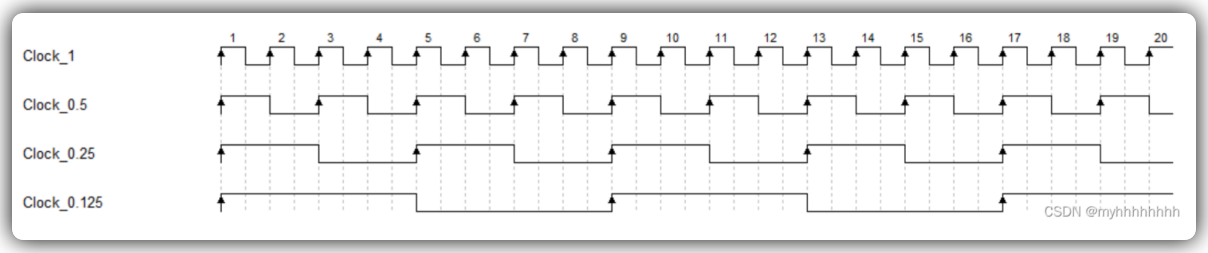
- 实现四分频电路，满足50%占空比。



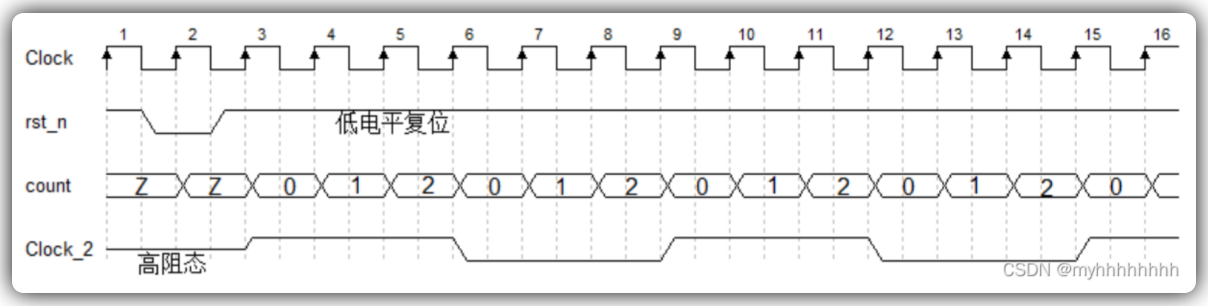
- 实现任意偶数分频，满足50%占空比。
- 实现任意偶数分频，不需要满足占空比。

2. 偶数分频电路原理

- 寄存器级联法
    - 寄存器级联法能实现 2^{N} 的偶数分频，具体是采用寄存器结构的电路，每当时钟上升沿到来的时候输出结果进行翻转，以此来实现偶数分频。



- 计数器法
  - 计数器法可以实现任意偶数分频，以六分频为例，电路需要实现的是：计数器从0开始计数至2，当clock上升沿取到clock=2时，输出clock进行翻转



3. RTL:寄存器级联法分频电路

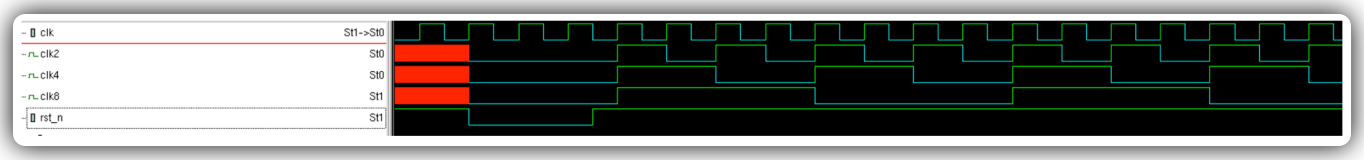
```
module even_divide(
   input logic clk,
   input logic rst_n,
   output logic clk_2,
   output logic clk_4,
   output logic clk_8
);

   always@(posedge clk or negedge rst_n) begin
      if(!rst_n) begin
         clk_2 <= 1'b0;
      end else begin
         clk_2 <= ~clk_2;
      end
   end

   always@(posedge clk_2 or negedge rst_n) begin
      if(!rst_n) begin
         clk_4 <= 1'b0;
      end else begin
         clk_4 <= ~clk_4;
      end
   end

   always@(posedge clk_4 or negedge rst_n) begin
      if(!rst_n) begin
         clk_8 <= 1'b0;
      end else begin
         clk_8 <= ~clk_8;
      end
   end

endmodule
```



4. RTL:计数器法分频电路

- N: 分频系数

```
module even_divide #(
    parameter int DIV_FACTOR = 2
)(
    input logic clk,
    input logic rst_n,
    output logic clk_out
);

    logic [31:0] counter;

    always@(posedge clk or negedge rst_n) begin
        if(!rst_n | counter == (DIV_FACTOR>>1) - 1) begin
            counter <= 32'd0;
        end else begin
            counter <= counter + 1;
        end
    end

    assign clk_out = (counter == (DIV_FACTOR>>1) - 1) ? ~clk_out : clk_out;

endmodule
```

